

PROFILE-BASED MATCHING SYSTEM

Intelligent Hybrid Recommendation System • NLP • Personality Compatibility • Adaptive Feedback

Student	Suzain Mudabbir Sayyad	University	Sanjay Ghodawat University, Kolhapur
Domain	Machine Learning / Recommendation Systems	Project Type	Major Project 1

1. Objective & Dataset

Build an end-to-end profile recommendation system that combines semantic profile similarity, MBTI personality compatibility, location compatibility and behavioral feedback to generate personalized Top-5 recommendations.

200 synthetic profiles	8 profile information fields	Top 5 recommendations per user	500 simulated feedback interactions
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2. Data Preparation & EDA

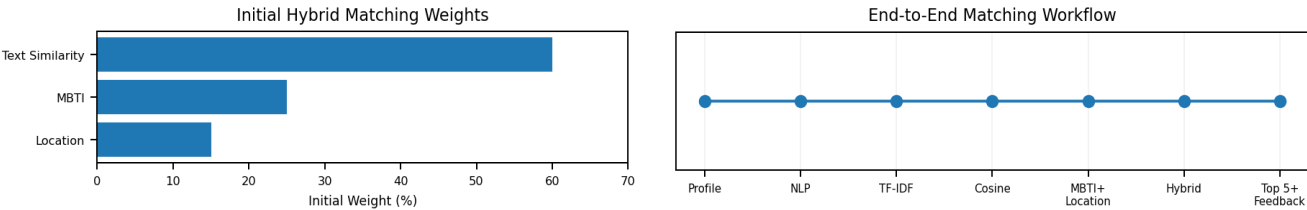
Text preprocessing Lowercase, punctuation removal, tokenization, English stop-word removal and lemmatization using NLTK.	Profile fields About Me, Professional Summary, Skills, Professional Goal, Interests, MBTI Personality Type and Location are used.
Combined text About Me + Professional Summary + Skills + Interests are combined before vectorization.	Similarity matrix TF-IDF vectors are compared using cosine similarity to obtain semantic profile similarity.

3. Hybrid Matching Model

The initial model combines three compatibility components. Text similarity has the highest influence, while personality and location provide complementary matching signals.

TEXT SIMILARITY 60%	MBTI COMPATIBILITY 25%	LOCATION COMPATIBILITY 15%
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$$\text{Compatibility Score} = (\text{Text Similarity} \times 0.60) + (\text{MBTI Match} \times 0.25) + (\text{Location Match} \times 0.15)$$



4. Adaptive Feedback Learning

Accept = 1 and Reject = 0 feedback is collected. Logistic Regression learns from Text Similarity, MBTI Score and Location Score ; learned coefficients are normalized into adaptive weights.	Recommendation output • Ranks candidates by compatibility score. • Returns Top 5 while excluding the selected user. • Compares original and adaptive recommendations. • Tracks acceptance and component-level feedback.
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5. Deployment & Final Output

Deployment	Output
Streamlit Dashboard • Existing User • New Profile • Upload Profiles • Model Analysis	matching_system.pkl • final_profiles.csv • adaptive_weights.csv
Generate personalized recommendations and upload-profile recommendations.	Top-5 recommendation CSV • feedback data • adaptive-weight analysis
Technology stack: Python, NLTK, Pandas, NumPy, Scikit-learn, Joblib, Streamlit, Matplotlib	Interactive model analysis and recommendation workflow

6. Conclusion

The completed system integrates NLP-based semantic matching with personality and location compatibility, then extends the scoring process with feedback-driven adaptive learning. The Streamlit application exposes the complete workflow for generating and analyzing personalized Top-5 recommendations.